

Mid Semester Examination, September 2023

Course Code: EIMTC04/ITMTC04/INMTC04

Course title: Optimization principles and techniques

Time: 3 hours

Maximum Marks. 50

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO																																				
1a	One kind of cake requires 200 g of flour and 25g of fat, and another kind of cake requires 100 g of flour and 50 g of fat Find the maximum number of cakes that can be made from 5 kg of flour and 1 kg of fat assuming that there is no shortage of the other ingredients, used in making the cakes. Form the LPP and solve it graphically.	5	CO1																																				
1b	Identify all basic feasible solutions for the following system. Also verify the number of degenerate and non-degenerate basic feasible solutions. $x_1 + x_2 + x_3 = 3$ $x_1 - x_2 + x_4 = 0, \quad x_1, x_2, x_3, x_4 \geq 0.$	5	CO1																																				
1c	Use two-phase simplex method to maximize Max $Z = 5x_1 + 3x_2$ subject to $2x_1 + x_2 \leq 1$ $x_1 + 4x_2 \geq 6, \quad x_1, x_2 \geq 0.$	5	CO1																																				
2a	Consider the following LPP: Max $Z = 4x_1 + 10x_2$ subject to $2x_1 + x_2 \leq 50$ $2x_1 + 5x_2 \leq 100$ $2x_1 + 3x_2 \leq 90, \quad x_1, x_2 \geq 0.$ The solution to the above LPP is $x_1 = 0, x_2 = 20$, and $Z = 200$. Write the dual of the above LPP. Also find the solution of the dual problem from the given solution of the primal problem.	5	CO2																																				
2b	Find the optimum quantity to be supplied from each shop to different warehouses at minimum total cost for the following cost matrix. <table><tr><td></td><td>D1</td><td>D2</td><td>D3</td><td>D4</td><td>D5</td><td>Supply</td></tr><tr><td>O1</td><td>20</td><td>18</td><td>18</td><td>21</td><td>19</td><td>100</td></tr><tr><td>O2</td><td>21</td><td>22</td><td>23</td><td>20</td><td>24</td><td>125</td></tr><tr><td>O3</td><td>18</td><td>19</td><td>21</td><td>18</td><td>19</td><td>175</td></tr><tr><td>Demand</td><td>60</td><td>80</td><td>85</td><td>105</td><td>70</td><td></td></tr></table>		D1	D2	D3	D4	D5	Supply	O1	20	18	18	21	19	100	O2	21	22	23	20	24	125	O3	18	19	21	18	19	175	Demand	60	80	85	105	70		5	CO2	
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2c	We have to assign five jobs, I, II, III, IV, V to five workers A, B, C, D, and E. The time taken by different workers (in hours) in completing different jobs is given below: <table><tr><td></td><td>I</td><td>II</td><td>III</td><td>IV</td><td>V</td></tr><tr><td>A</td><td>160</td><td>130</td><td>175</td><td>190</td><td>200</td></tr><tr><td>B</td><td>135</td><td>120</td><td>130</td><td>160</td><td>175</td></tr><tr><td>C</td><td>140</td><td>110</td><td>155</td><td>170</td><td>185</td></tr><tr><td>D</td><td>50</td><td>50</td><td>80</td><td>80</td><td>110</td></tr><tr><td>E</td><td>55</td><td>35</td><td>70</td><td>80</td><td>105</td></tr></table> Find the minimum time (in hours) taken by the workers to complete all jobs.		I	II	III	IV	V	A	160	130	175	190	200	B	135	120	130	160	175	C	140	110	155	170	185	D	50	50	80	80	110	E	55	35	70	80	105	5	CO2
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D	50	50	80	80	110																																		
E	55	35	70	80	105																																		

3a	Define a strategy? What do you mean by pure strategy and mixed strategy? Give examples. Define saddle points. Also describe the Maximin-Minimax principle with example for a two person zero sum game.	5	CO3
3b	Solve the following 2X3 game graphically <i>Player B</i> 1 3 11 7 <i>Player A</i> 8 5 2 6	5	CO3
3c	Using the principle of dominance, solve the following game problem <i>Player B</i> 4 6 5 10 6 7 8 5 9 10 <i>Player A</i> 8 9 11 10 9 6 4 10 6 4	5	CO3
4a	Solve the following linear programming problem using dual simplex method. Min $Z = 10x_1 + 6x_2 + 2x_3$ subject to $-x_1 + x_2 + x_3 \geq 1$ $3x_1 + x_2 - x_3 \geq 2, \quad x_1, x_2, x_3 \geq 0.$	5	CO4
4b	Solve the integer programming problem given as Max $Z = 3x_2$ subject to $3x_1 + 2x_2 \leq 7$ $x_1 - x_2 \geq -2, \quad x_1 \geq 0, x_2 \geq 0, \quad x_1, x_2 \text{ is an integer}$	5	CO4
4c	Solve the integer programming problem given as Max $Z = 3x + 2y$ subject to $x + y \leq 6$ $5x + 2y \leq 20, \quad x \geq 0, y \geq 0, \quad y \text{ is an integer}$	5	CO4
5a	Solve the non linear programming problem by forming a bordered hessian matrix: Optimize $Z = x_1^2 + x_2^2 + x_3^2$ subject to $x_1 + x_2 + 3x_3 = 2$ $5x_1 + 2x_2 + x_3 = 5$ $x_1, x_2, x_3 \geq 0$	5	CO5
5b	Solve the following NLPP: Optimize $Z = -x_1^2 - x_2^2 - x_3^2 + 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 1$ $2x_1 + 3x_2 \leq 6$ $x_1, x_2, x_3 \geq 0$	5	CO5
5c	Define convex programming problem. Check if the following problems are convex programming problems or not: i) Max $Z = 4x_1 + 3x_2$ subject to $x_1^2 + x_2^2 \leq 1$ $(x_1 - 1)^2 + x_2^2 \leq 1$ ii) Min $Z = x_1 + x_2$ subject to $-x_1x_2 + 1 \leq 0$ $x_1, x_2 \geq 0$	5	CO5